

Vol 9 Chapter 10 — Strongly Correlated Electron Systems

Keeper (author pass — deep math/physics revision)

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Chapter 10 — Strongly Correlated Electron Systems

Level 1 — one sentence

Strongly correlated electron systems — Mott insulators, heavy fermions, fractional quantum Hall liquids, doped cuprates — are systems where electron-electron interactions dominate over kinetic energy, breaking single-particle band-theory descriptions and requiring full many-body treatment; in BST these are substrate K-type configurations where K-type-K-type couplings can't be approximated by mean-field.

Level 2 — graduate-physicist precision

10.1 The Hubbard model

The simplest strongly-correlated model: electrons on a lattice with on-site repulsion U and hopping t :

$$H = -t \sum_{\langle i,j \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

Two limits: - $U/t \ll 1$: nearly free electrons $\rightarrow$ band metal - $U/t \gg 1$: strong correlations $\rightarrow$ Mott insulator (no double occupancy)

At half-filling, large U/t : insulating despite half-filled band. Conventional band theory predicts metal — wrong because correlation freezes electrons in place.

Mott metal-insulator transition: occurs at critical $U_c/t \sim \text{few}$. Driven by correlation, not band-filling.

10.2 Heavy fermion materials

Compounds containing f-electron elements (Ce, U, Yb): conduction electrons hybridize with localized f-electrons via Kondo coupling. At low T, Kondo screening + RKKY interactions produce effective heavy quasiparticles with $m^* \sim 100 - 1000 m_e$.

Examples: CeAl_3 , UPt_3 , CeCoIn_5 . Show exotic superconductivity, non-Fermi-liquid metals, quantum critical points.

10.3 Fractional quantum Hall liquid

(Discussed Ch 5 Section 5.2): 2D electron gas in strong B field at fractional filling $\nu = 1/3, 2/5, \dots$ forms a topologically-ordered liquid with fractionally-charged quasiparticles and anyonic statistics.

Laughlin wave function $\Psi_{1/3} \propto \prod (z_i - z_j)^3 \exp(-\sum |z|^2/4)$ for $\nu = 1/3$. Beautifully simple.

10.4 High- T_c cuprates as doped Mott insulators

Cuprates (Ch 4) are doped Mott insulators: parent compound (e.g., La_2CuO_4) is antiferromagnetic Mott insulator; doping with holes (Sr substitution) or electrons gives superconductor + various exotic phases.

The cuprate phase diagram (T vs doping) shows: - Antiferromagnet near zero doping - Superconductor in dome around optimal doping - Pseudogap above T_c in underdoped region - Strange metal in overdoped region

40+ years of work; consensus mechanism remains elusive. BST predicts substrate K-type origin (Ch 4, Task #88).

10.5 Quantum spin liquids (Ch 11 preview)

A spin liquid is a Mott insulator with no magnetic ordering even at $T = 0$ — strong correlation prevents single-particle picture; geometric frustration prevents any classical-order ground state. Topological structure: fractionalized spinon excitations + emergent gauge fields.

10.6 Numerical methods

Strongly correlated systems are computationally hard: - DMRG (density matrix renormalization group): exact for 1D ground states - QMC (quantum Monte Carlo): exact for sign-problem-free models (boson, half-filled, etc.) - DMFT (dynamical mean-field theory): self-consistent local-correlations approach - Exact diagonalization: only small systems (up to ~24 sites)

10.7 BST predictions

Task #105 (iron pnictide T_c), Task #106 (topological insulator classification), Task #107 (spin liquid order parameters), Task #108 (QHE plateau precision) — BST has filed predictions for various strongly-correlated phenomena. Cuprate Task #88 is the most prominent.

10.8 K-audit anchors

- Ch 4: cuprate superconductivity (strongly correlated)
- Ch 5: topological phases (overlap with FQHE)
- Ch 11: spin liquids
- Tasks #88, #105-108: BST predictions

Level 3 — 5th-grader accessibility

Strongly correlated = electrons interact with each other so strongly that single-particle band theory fails. Mott insulators: insulating despite half-filled band (correlation freezes electrons). Heavy fermions: f-electron compounds with effective electron mass 100-1000× free electron. Fractional QHE: 2D electrons in strong B field forming exotic quasi-particles with charge $e/3$. Cuprate high- T_c : doped Mott insulators; ~40 years of work and still not fully understood. BST predicts mechanism via substrate K-type structure.

What comes next

Chapter 11 develops spin liquids.

Where to look this up

- Fulde, *Electron Correlations in Molecules and Solids*
- Stewart, “Heavy fermion systems” Rev Mod Phys
- BST: Tasks #88, #105-108